



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

CE F231 Fluid Mechanics

Sayantan Chakraborty, Ph.D., AM ASCE, MIE
sayantan.chakraborty@pilani.bits-pilani.ac.in

November 16, 2022



Lecture 26: Orifices and Mouthpieces

Learning Outcome

- What are orifices and mouthpieces and their classifications?
- Estimation of flow through orifices (small, large, freely discharging, fully and partially submerged)
- Determination of hydraulic coefficients (C_d , C_c , and C_v)
- Time of emptying tanks through orifices
- Estimation of flow through different mouthpieces

Orifices

- Orifice → small aperture on side or bottom of a tank through which the fluid passes
- Thickness of orifice in direction of flow is very small in comparison to its other dimensions
- Orifice usually has a sharp edge

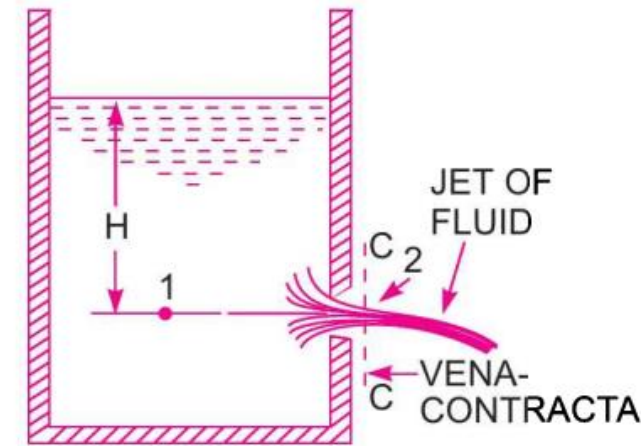


Fig. 7.1 Tank with an orifice.



Orifices: Classification

➤ Small or large orifice

For depth, check
slide 13,
 $\text{depth} = H_2 - H_1$

- Head of liquid $> 5 \times$ depth of orifice $\rightarrow$ small orifice
- Head of liquid $< 5 \times$ depth of orifice $\rightarrow$ large orifice

➤ Classification based on shape of cross-sectional area

- Circular, triangular, rectangular, square, etc.

➤ Classification based on shape of upstream edge

- Sharp-edged, bell-mouthed

➤ Classification based on nature of discharge

- Freely discharging, fully submerged, and partially submerged

Flow through orifice (small orifice - freely discharging)

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$z_1 = z_2$$

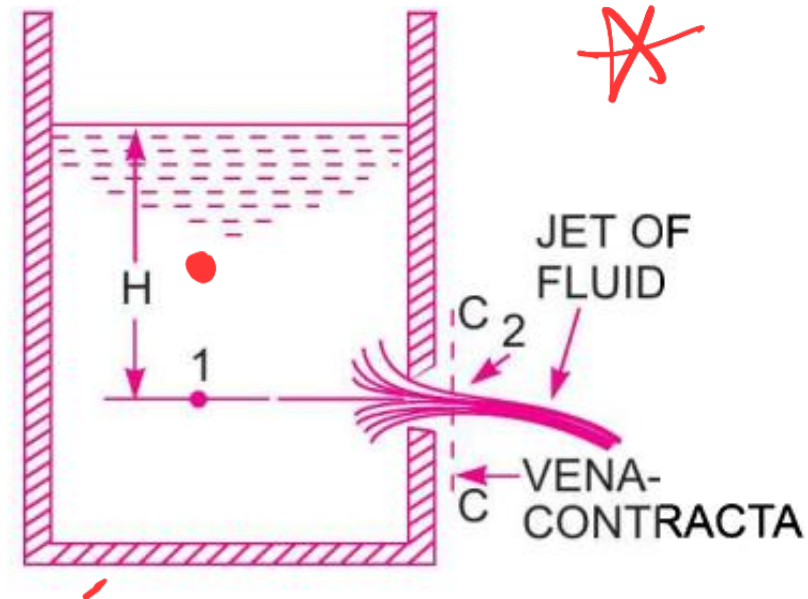
$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} = \frac{p_2}{\rho g} + \frac{v_2^2}{2g}$$

$$\frac{p_1}{\rho g} = H$$

$$\frac{p_2}{\rho g} = 0 \text{ (atmospheric pressure)}$$

$$H + 0 = 0 + \frac{v_2^2}{2g}$$

$$v_2 = \sqrt{2gH}$$



At vena contracta

- Flow is uniform
- Jet is horizontal

Hydraulic coefficients

- Coefficient of velocity (C_v)

$$C_v = \frac{\text{Actual velocity of jet at vena - contracta}}{\text{Theoretical velocity}}$$

$$= \frac{V}{\sqrt{2gH}}, \text{ where } V = \text{actual velocity, } \sqrt{2gH} = \text{Theoretical velocity}$$

- The value of C_v varies from 0.95 to 0.99 for different orifices
- Value of $C_v = 0.98$ for sharp-edged orifice

Hydraulic coefficients

- Coefficient of contraction (C_c)

$$C_c = \frac{\text{area of jet at vena-contracta}}{\text{area of orifice}}$$
$$= \frac{a_c}{a}$$

- The value of C_c varies from 0.61 to 0.69 for different orifices
- Value of C_c can be taken as 0.64

Hydraulic coefficients

- Coefficient of discharge (C_d)

$$C_d = \frac{Q}{Q_{th}} = \frac{\text{Actual velocity} \times \text{Actual area}}{\text{Theoretical velocity} \times \text{Theoretical area}}$$

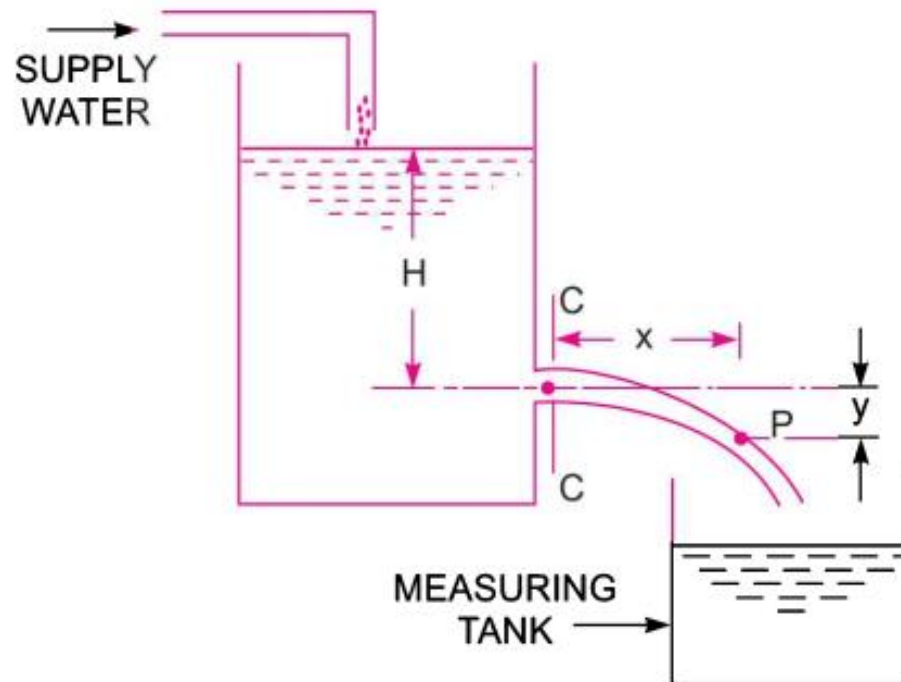
$$= \frac{\text{Actual velocity}}{\text{Theoretical velocity}} \times \frac{\text{Actual area}}{\text{Theoretical area}}$$

$$C_d = C_v \times C_c$$

- The value of C_d varies from 0.61 to 0.65 for different orifices
- Value of C_d can be taken as 0.62

Determination of hydraulic coefficients (experimental)

➤ Coefficient of discharge (C_d)



$$Q = \frac{\text{Area of measuring tank} \times \text{Height of water in measuring tank}}{\text{Time } (t)}$$

$$C_d = \frac{Q}{a \times \sqrt{2gH}}$$

$$\text{theoretical discharge} = \text{area of orifice} \times \sqrt{2gH}$$

Determination of hydraulic coefficients (experimental)

➤ Coefficient of velocity (C_v)

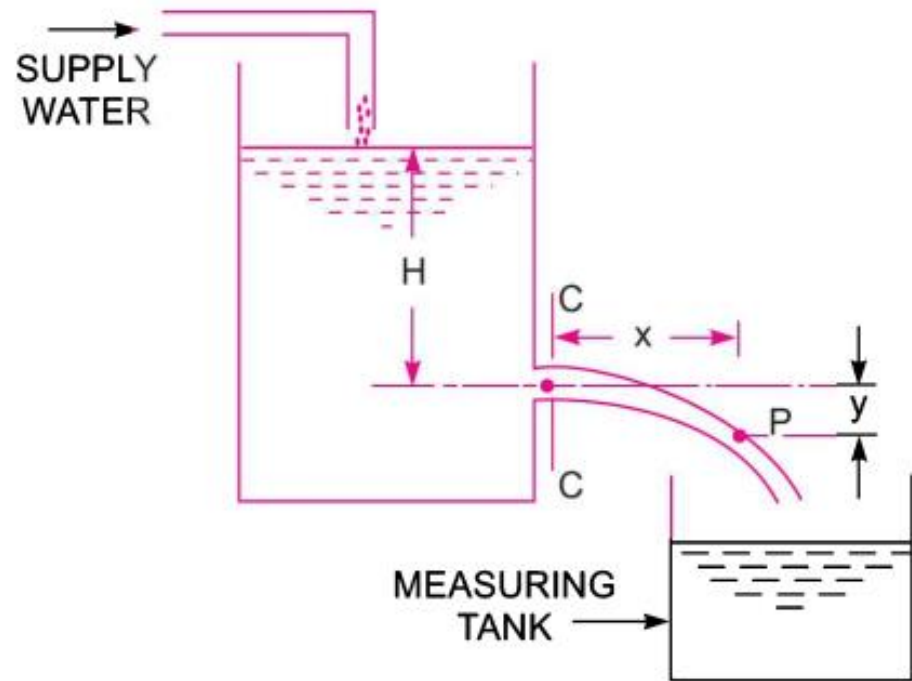
$$x = V \times t$$

$$y = \frac{1}{2} g t^2$$

$$y = \frac{1}{2} g \times \frac{x^2}{V^2}$$

$$V^2 = \frac{gx^2}{2y}$$

$$V = \sqrt{\frac{gx^2}{2y}}$$



$$\begin{aligned} \text{Co-efficient of velocity, } C_v &= \frac{V}{V_{th}} = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}} \\ &= \frac{x}{\sqrt{4yH}} \end{aligned}$$

Determination of hydraulic coefficients (experimental)

- Coefficient of contraction (C_c)

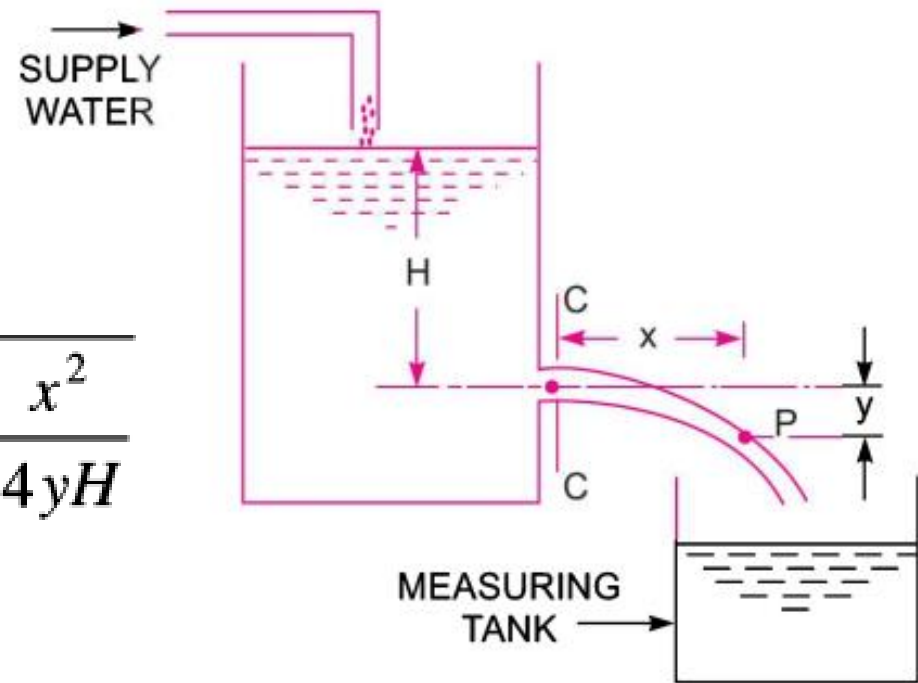
$$C_d = \frac{Q}{a \times \sqrt{2gH}}$$

$$C_v = \frac{V}{V_{th}} = \sqrt{\frac{gx^2}{2y}} \times \frac{1}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$$

$$= \frac{x}{\sqrt{4yH}}$$

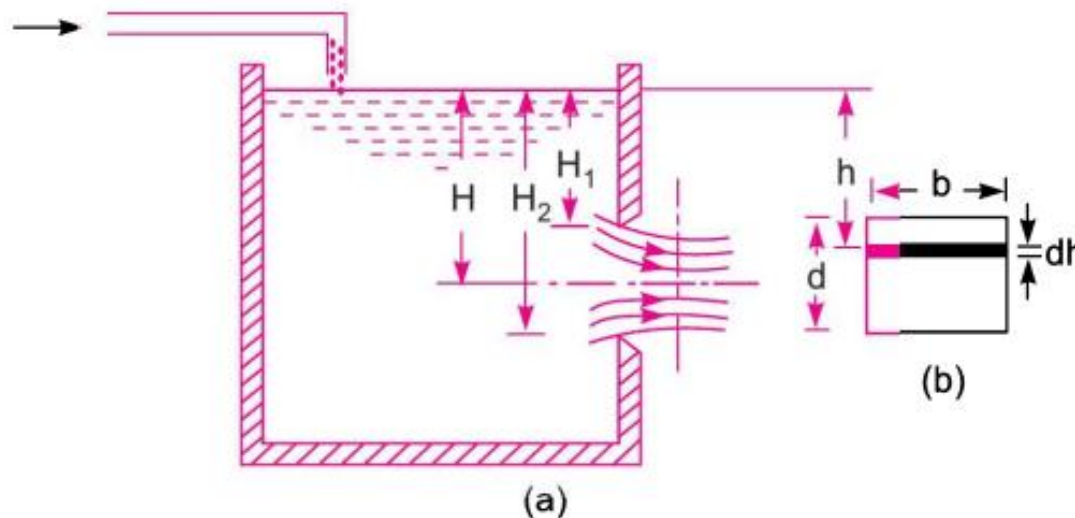
$$C_d = C_v \times C_c$$

$$C_c = \frac{C_d}{C_v}$$

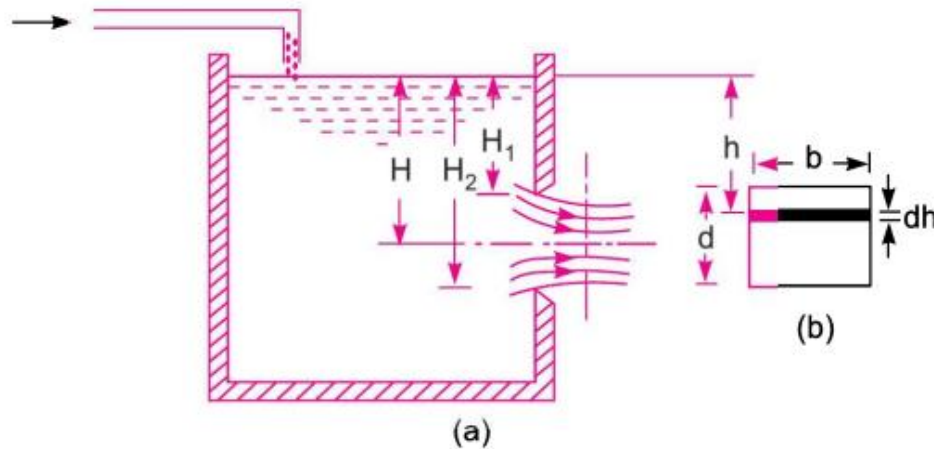


Flow through large orifices

- Head of liquid $< 5 \times$ depth of orifice
- For small orifices $\rightarrow$ velocity in entire cross-section of jet is assumed to be constant
- For large orifices $\rightarrow$ velocity in entire cross-section cannot be assumed to be constant



Flow through large rectangular orifices (freely discharging)



H_1 = height of liquid above top edge of orifice
 H_2 = height of liquid above bottom edge of orifice
 b = breadth of orifice
 d = depth of orifice = $H_2 - H_1$
 C_d = co-efficient of discharge.

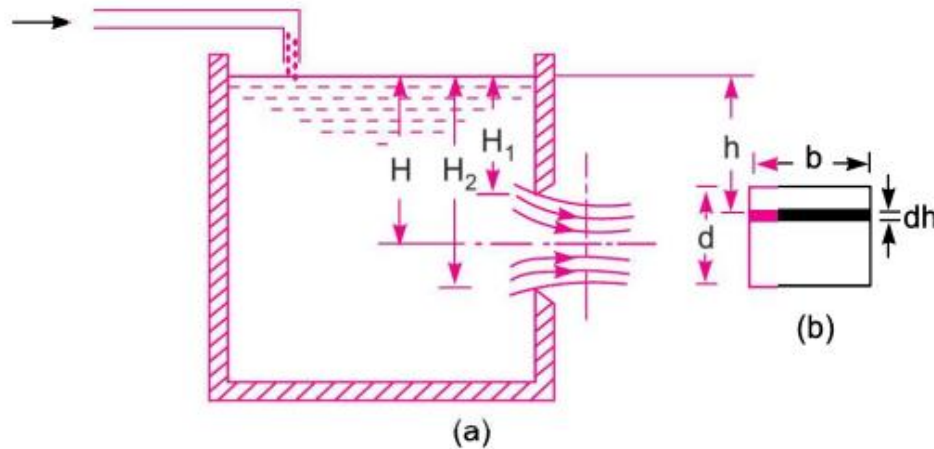
∴ Area of strip = $b \times dh$

and theoretical velocity of water through strip = $\sqrt{2gh}$.

∴ Discharge through elementary strip is given

$$\begin{aligned}
 dQ &= C_d \times \text{Area of strip} \times \text{Velocity} \\
 &= C_d \times b \times dh \times \sqrt{2gh} = C_d b \times \sqrt{2gh} \, dh
 \end{aligned}$$

Flow through large rectangular orifices (freely discharging)



H_1 = height of liquid above top edge of orifice

H_2 = height of liquid above bottom edge of orifice

b = breadth of orifice

d = depth of orifice = $H_2 - H_1$

C_d = co-efficient of discharge.

$$Q = \int_{H_1}^{H_2} C_d \times b \times \sqrt{2gh} \, dh$$

$$= C_d \times b \times \sqrt{2g} \int_{H_1}^{H_2} \sqrt{h} \, dh = C_d \times b \times \sqrt{2g} \left[\frac{h^{3/2}}{3/2} \right]_{H_1}^{H_2}$$

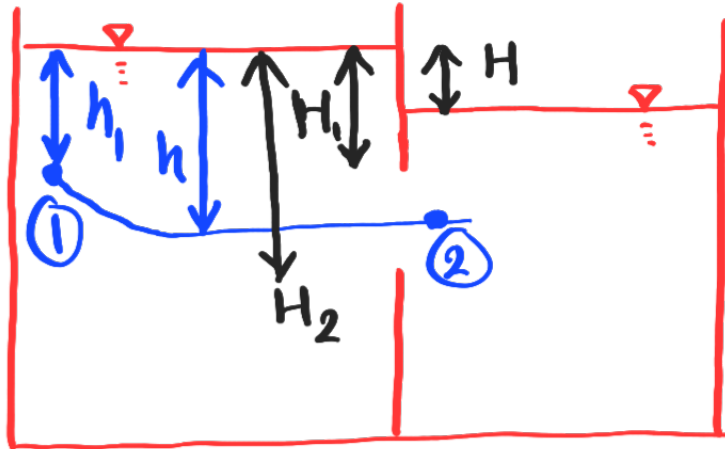
$$= \frac{2}{3} C_d \times b \sqrt{2g} [H_2^{3/2} - H_1^{3/2}]$$

Discharge through fully submerged orifice

innovate

achieve

lead



$$h - h_1 + h_1 + 0$$

$$= 0 + h - H + \frac{v^2}{2g}$$

$$\Rightarrow v = \sqrt{2gH}$$

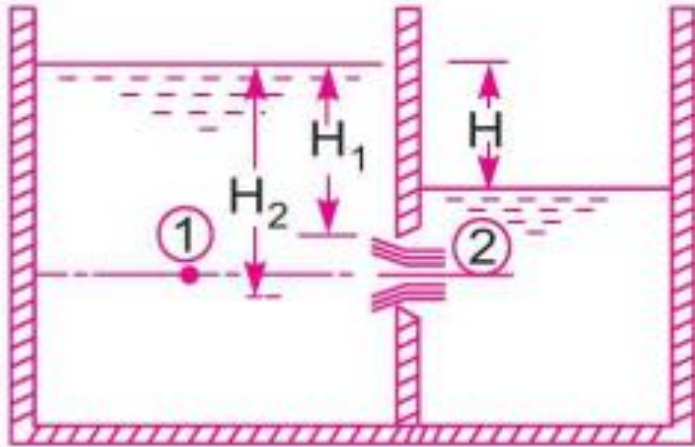
$$(H_1 \leq h \leq H_2)$$

Discharge through fully submerged orifice

innovate

achieve

lead



H_1 = Height of water above the top of the orifice on the upstream side,

H_2 = Height of water above the bottom of the orifice,

H = Difference in water level,

b = Width of orifice,

C_d = Co-efficient of discharge.

$\therefore$

$$\frac{V_2^2}{2g} = H$$

$\therefore$

$$V_2 = \sqrt{2gH}$$

Area of orifice

$$= b \times (H_2 - H_1)$$

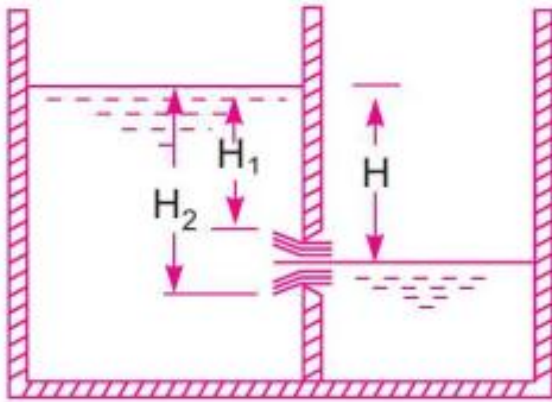
$\therefore$ Discharge through orifice = $C_d \times \text{Area} \times \text{Velocity}$

$$= C_d \times b (H_2 - H_1) \times \sqrt{2gH}$$

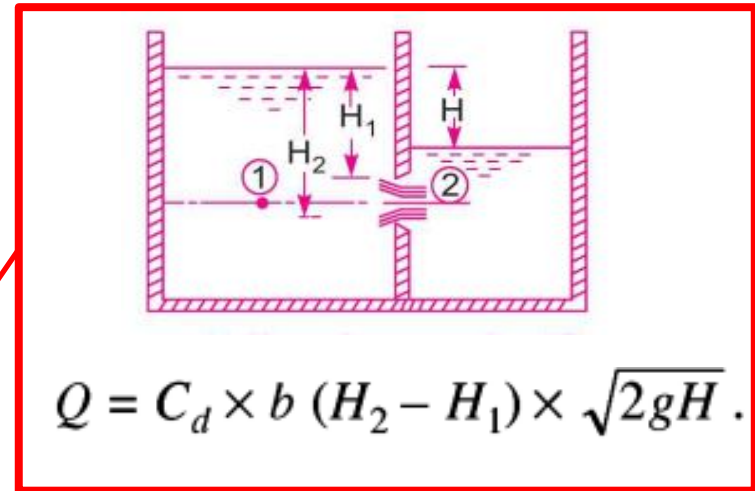
$\therefore$

$$Q = C_d \times b (H_2 - H_1) \times \sqrt{2gH}.$$

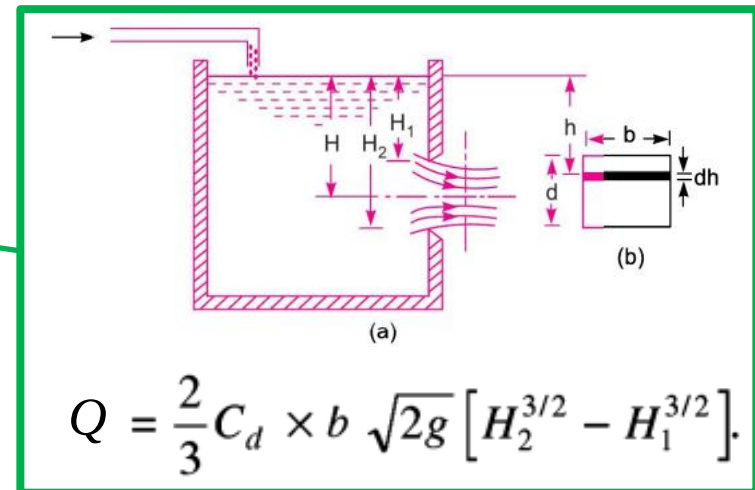
Discharge through partially submerged orifice



Fully submerged



Freely discharging



$$Q_1 = C_d \times b \times (H_2 - H) \times \sqrt{2gH}$$

$$Q_2 = \frac{2}{3} C_d \times b \times \sqrt{2g} [H^{3/2} - H_1^{3/2}]$$

Total discharge

$$Q = Q_1 + Q_2$$

$$Q = \frac{2}{3} C_d \times b \sqrt{2g} [H_2^{3/2} - H_1^{3/2}]$$

Time of emptying cylindrical tank through orifice at bottom of tank

Let A = Area of the tank

a = Area of the orifice

H_1 = Initial height of the liquid

H_2 = Final height of the liquid

T = Time in seconds for the liquid to fall from H_1 to H_2 .

Let at any time, the height of liquid from orifice is h and let the liquid surface fall by a small height dh in time dT . Then

Volume of liquid leaving the tank in time, $dT = A \times dh$

Also the theoretical velocity through orifice, $V = \sqrt{2gh}$

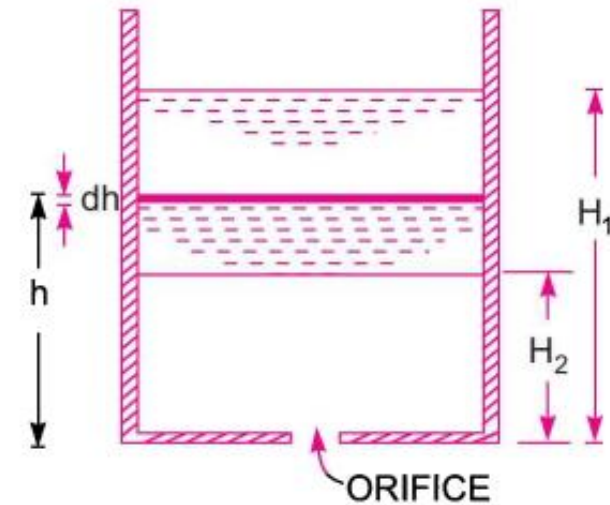
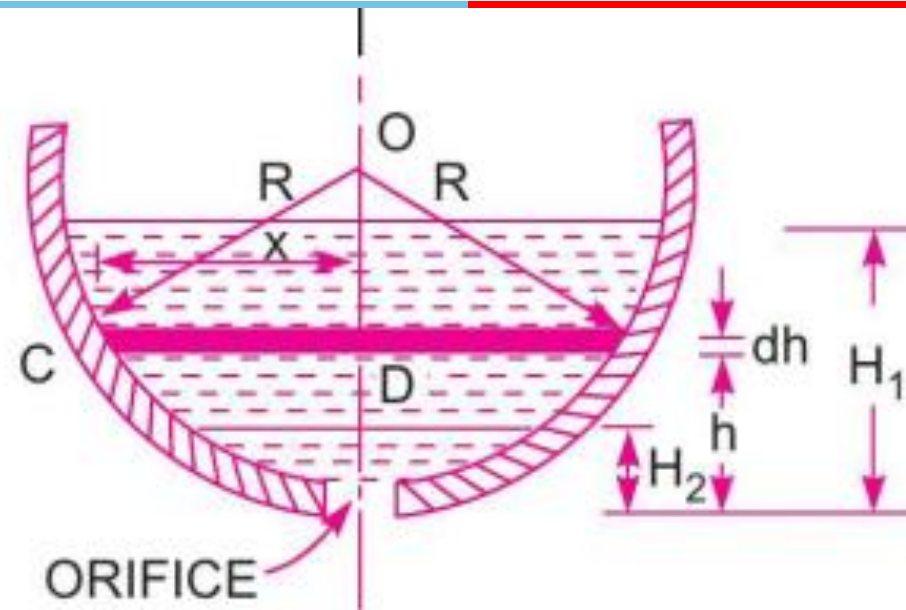


Fig. 7.9. (a)

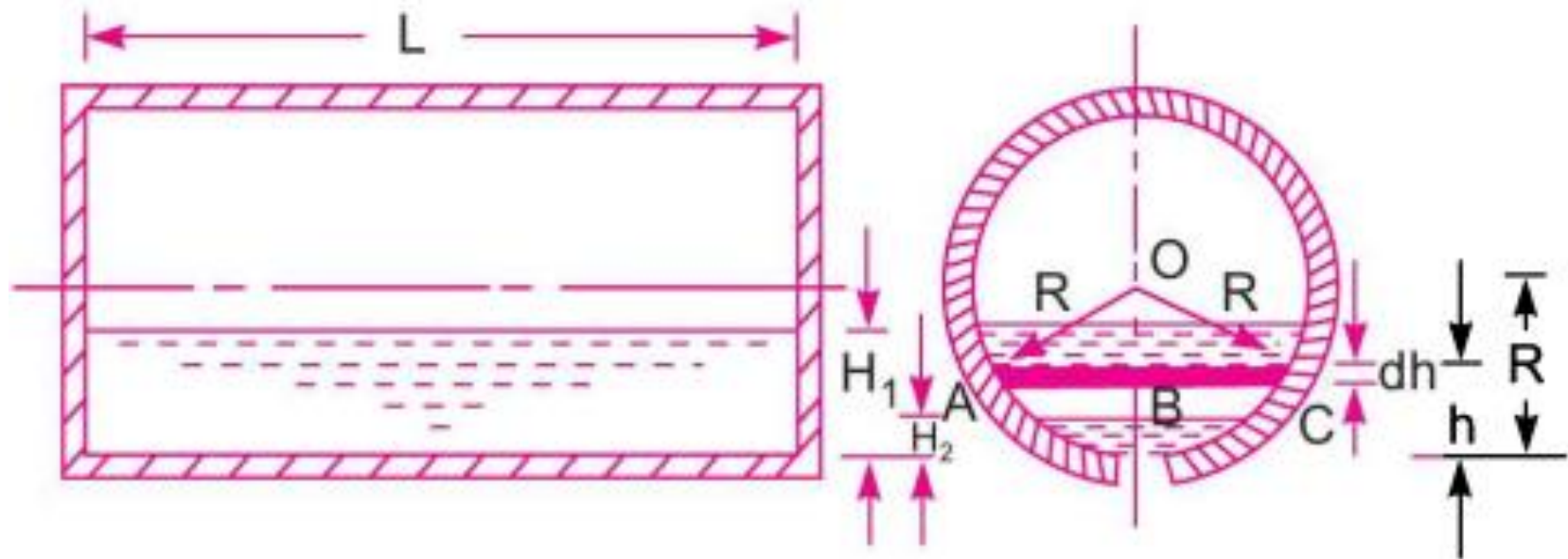
$$T = \frac{2A \left[\sqrt{H_1} - \sqrt{H_2} \right]}{C_d \cdot a \cdot \sqrt{2g}}$$

Time of emptying hemispherical tank through orifice at bottom of tank



$$T = \frac{\pi}{C_d \times a \times \sqrt{2g}} \left[\frac{4}{3} R (H_1^{3/2} - H_2^{3/2}) - \frac{2}{5} (H_1^{5/2} - H_2^{5/2}) \right]$$

Time of emptying circular horizontal tank



$$T = \frac{4L}{3C_d \times a \times \sqrt{2g}} \left[(2R - H_2)^{3/2} - (2R - H_1)^{3/2} \right]$$



THANK YOU